

1.Display emp_id, emp_name, dept_id, and salary for employees whose salary is greater than the average salary of their own department. Do not use JOIN.

```
select emp_id,emp_name,dept_id,salary from employees e where  
salary>(select avg(salary) from employees where dept_id=e.dept_id);
```

2.Display the highest-paid employee or employees from each department. If multiple employees in the same department share the highest salary, return all of them

```
select emp_id,emp_name,dept_id,salary from employees e where  
salary=(select max(salary) from employees where dept_id=e.dept_id);
```

3.Find the second-highest distinct salary in the company without using LIMIT, OFFSET, or JOIN

```
select distinct max(salary) from employees where salary<(select max(salary)  
from employees);
```

4.Display the employee name and salary of every employee whose salary equals the second-highest distinct salary in the company.

```
select emp_name,salary from employees where salary=(select max(salary)  
from employees where salary<(select max(salary) from employees));
```

5.Display employees whose salary is greater than the salary of their own manager. Employees without a manager should not appear. Do not use a self JOIN.

```
select e.emp_id,e.emp_name,e.salary from employees e where  
e.salary>(select m.salary from employees m where  
m.emp_id=e.manager_id);
```

6.Display employees who are managers and whose salary is less than at least one employee reporting directly to them.

```
select e.emp_id, e.emp_name, e.salary from employees e where exists  
(select 1 from employees x where x.manager_id = e.emp_id and x.salary >  
e.salary);
```

7.Display the department IDs whose average employee salary is greater than the overall average salary of all employees.

```
SELECT dept_id FROM employees GROUP BY dept_id HAVING AVG(salary) >  
(SELECT AVG(salary) FROM employees);
```

8.Display all employees who belong to departments whose departmental average salary is greater than the overall company average salary. Do not use JOIN.

```
SELECT emp_id, emp_name, dept_id, salary FROM employees WHERE  
dept_id IN (SELECT dept_id FROM employees GROUP BY dept_id HAVING  
AVG(salary) > (SELECT AVG(salary) FROM employees));
```

9.Display employees whose salary is greater than the salary of every employee in the HR department. The department ID must be obtained using a subquery from the departments table.

```
select emp_id, emp_name, salary from employees where salary > all (select salary from employees where dept_id = (select dept_id from departments where dept_name = 'HR'));
```

10.Display employees whose salary is greater than at least one employee working in the Research department. Obtain the Research department ID through a subquery.

```
select emp_id, emp_name, salary from employees where salary > any (select salary from employees where dept_id = (select dept_id from departments where dept_name = 'Research'));
```

11.Display dept_id and dept_name for departments that do not have any project listed in the projects table. Solve using a subquery only.

```
select dept_id, dept_name from departments d where not exists (select 1 from projects p where p.dept_id = d.dept_id);
```

12.Display all employees who belong to a department that has more than one project. Do not use JOIN

```
select emp_id, emp_name, dept_id, salary from employees e where (select count(*) from projects p where p.dept_id = e.dept_id) > 1;
```

13.Display all employees who work in the department that owns the project having the maximum budget. If multiple projects tie for maximum budget, employees from all matching departments should be returned.

```
select emp_id, emp_name, dept_id, salary from employees where dept_id in  
(select dept_id from projects where budget = (select max(budget) from  
projects));
```

14.Display the employee or employees whose salary has the smallest absolute difference from the company average salary. Do not use ORDER BY ... LIMIT and do not use JOIN.

```
select emp_id, emp_name, salary from employees where abs(salary - (select  
avg(salary) from employees)) = (select min(abs(salary - (select avg(salary)  
from employees))) from employees);
```

15.Display department IDs where the maximum employee salary in that department is greater than the salary of every employee in the same department who is acting as a manager of at least one other employee.

```
select e.dept_id from employees e group by e.dept_id having max(e.salary)  
> all (select m.salary from employees m where m.dept_id = e.dept_id and  
exists (select 1 from employees x where x.manager_id = m.emp_id));
```